Bachelor of Computer

Applications Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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Experiment No.: 1

Importing Business Datasets from CSV Flat Files and Connecting with Other Sources into Power BI

GitHub Link: https://github.com/Lakhan07AU/Bussiness-AnalyticsProjects/tree/main/Experiment-1

Problem Statement

As a Data Analyst in a retail company, sales, customer, and product data are received as separate flat files. Management needs a centralized, interactive reporting system that consolidates these sources so that order patterns can be analysed by product, customer and time. This experiment imports the AdventureWorks Sales, Customer and Product datasets into Power BI, links them into a single data model, and builds a one-page dashboard for business reporting.

Step 1: Data Collection

The following flat files (CSV) were used as the data sources for this experiment:

File	Rows	Key Columns
AdventureWorks_Sales_Data_2022.csv	29,481 order lines	OrderDate, StockDate, OrderNumber, ProductKey, CustomerKey, TerritoryKey, OrderLineItem, OrderQuantity
AdventureWorks_Customer_Lookup.csv	18,154 customers	CustomerKey, Prefix, FirstName, LastName, BirthDate, MaritalStatus, Gender, EmailAddress, AnnualIncome, TotalChildren, EducationLevel, Occupation, HomeOwner
AdventureWorks_Product_Lookup.csv	293 products	ProductKey, ProductSubcategoryKey, ProductSKU, ProductName, ModelName, ProductColor, ProductSize, ProductStyle, ProductCost, ProductPrice

The Sales dataset records order-level transactions for the period 1 January 2022 to 30 June 2022 (11,839 distinct orders across 102 distinct products), the Customer Lookup file holds demographic details for 18,154 customers, and the Product Lookup file holds pricing and attribute details for 293 products.

Step 2: Data Import into Power BI

- Opened Power BI Desktop and selected Get Data → Text/CSV.

- Imported AdventureWorks_Sales_Data_2022.csv, AdventureWorks_Customer_Lookup.csv and AdventureWorks_Product_Lookup.csv directly as flat-file sources.
- Each file was previewed in the navigator before loading, and column data types were reviewed at import time.
- All three tables were loaded into the Power BI data model as AdventureWorks Sales Data 2022, AdventureWorks Customer Lookup and AdventureWorks Product Lookup (a MySQL/other-source connection can equally be used for any one of these tables to satisfy a multi-source requirement, following the same Get Data workflow with the MySQL database connector instead of Text/CSV).

Step 3: Data Cleaning and Transformation

- Opened Power Query Editor for each table.
- Checked the datasets for missing values, duplicate rows and inconsistent entries.
- Verified column names and corrected data types, e.g. OrderDate and StockDate set to Date, OrderQuantity to Whole Number, ProductPrice/ProductCost to Decimal Number.
- Confirmed that the key columns ProductKey and CustomerKey were formatted consistently across all three tables so relationships could be created.
- Removed columns not required for this analysis and closed & applied the queries to load the cleaned tables into the model.

Step 4: Data Modeling

- Created a relationship between AdventureWorks Sales Data 2022[ProductKey] and AdventureWorks Product Lookup[ProductKey].
- Created a relationship between AdventureWorks Sales Data 2022[CustomerKey] and AdventureWorks Customer Lookup[CustomerKey].
- Both relationships are one-to-many (lookup table to sales table) with a single filter direction from the lookup tables to the Sales table.
- For date-based analysis, Power BI's automatic Date Hierarchy (Year → Quarter → Month → Day) built on the Sales table's OrderDate column was used directly, so a separate Calendar table was not required for this single-page dashboard.

Step 5: Measures Used in the Dashboard

The dashboard visuals use Power BI's built-in Sum aggregation on OrderQuantity and ProductPrice rather than custom DAX measures, since the analysis in this experiment is centred on order-quantity trends. The core aggregation used across the visuals is:

```
Sum of OrderQuantity  =  SUM('AdventureWorks Sales Data 2022'[OrderQuantity])
```

This single aggregation is reused as the Values field in the KPI card, bar chart, line chart and donut chart, while the scatter chart additionally uses Sum of ProductPrice from the Product Lookup table on its X-axis.

Step 6: Dashboard Development

A single-page Power BI report was built containing the following visuals:

Visual	Type	Fields Used
Total Quantity Sold	Card	Sum of OrderQuantity (Sales Data 2022)
Sales Quantity By Product	Clustered bar chart	Axis: ProductName (Product Lookup); Value: Sum of OrderQuantity
Sales Trend Over Time	Line chart	Axis: OrderDate Date Hierarchy (Year/Quarter/Month/Day); Value: Sum of OrderQuantity
Product Price vs Sales Quantity	Scatter chart	X: Sum of ProductPrice (Product Lookup); Y: Sum of OrderQuantity
Sales By Customer Gender	Donut chart	Legend: Gender (Customer Lookup); Value: Sum of OrderQuantity

Text-box headers were added above each visual (Total Quantity Sold, Sales Quantity By Product, Sales trend Over Time, Product Price vs Sales Quantity, Sales By Customer Gender) so the dashboard is self-explanatory without a separate legend page.

Step 7: Dashboard Formatting

- Arranged the card and four charts on a single 1920×1080 page in a clear grid layout.
- Added a descriptive text-box title above each visual instead of relying only on default visual titles.
- Kept a consistent Fluent-based theme across all visuals for a uniform colour palette.
- Sorted the bar and donut charts in descending order of Sum of OrderQuantity so the largest contributors are easiest to spot.

Step 8: Analysis and Insights

Based on the underlying data and the completed dashboard:

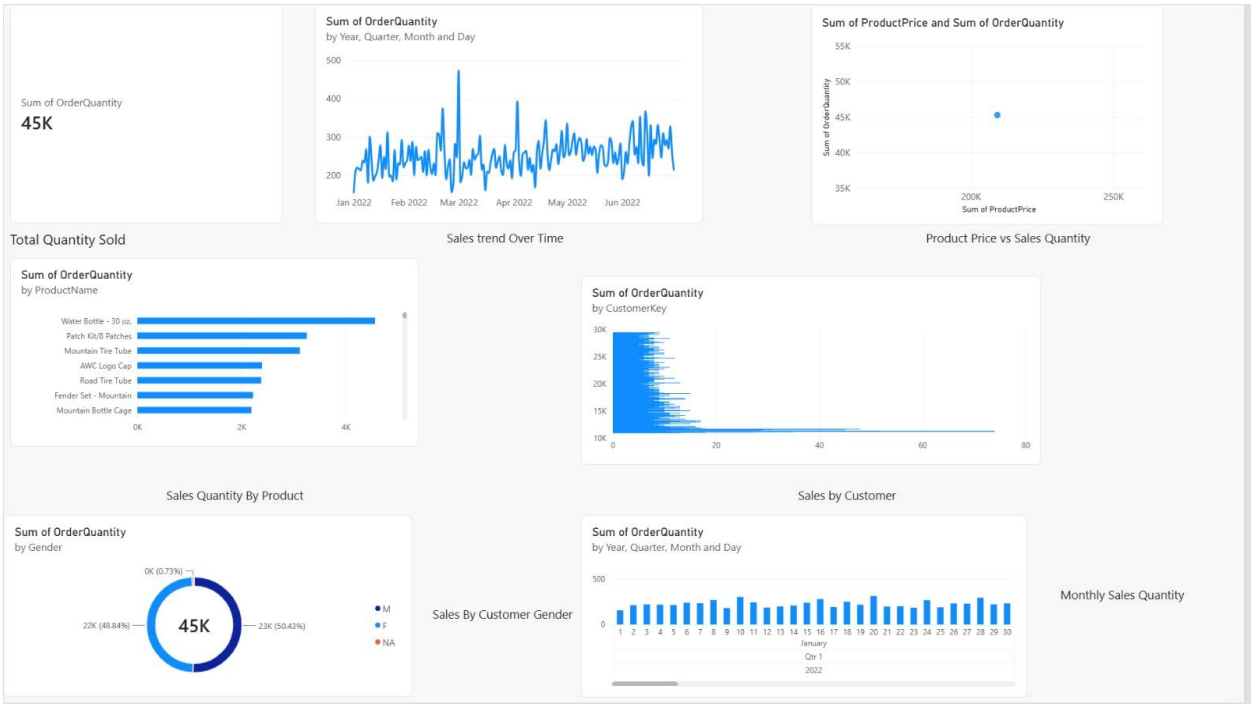
- Total quantity sold across the reporting period (Jan–Jun 2022) is 45,314 units, from 29,481 order lines across 11,839 distinct orders.
- The Water Bottle - 30 oz. is the top-selling item by quantity (4,565 units), followed by the Patch Kit/8 Patches (3,255), Mountain Tire Tube (3,121), AWC Logo Cap (2,394) and Road Tire Tube (2,379) — low-cost accessories dominate order volume.
- Order quantity is fairly evenly split by customer gender: Male customers account for 22,851 units and Female customers for 22,130 units.
- The monthly trend line shows order quantity rising steadily from 7,020 units in January to 8,260 units in June, indicating growing demand across the first half of 2022.

- The scatter chart shows a negative relationship between product price and order quantity, i.e. lower-priced products tend to sell in higher volumes than premium-priced items.

Step 9: Final Dashboard

After completing data import, cleaning, modeling and visualization, the final one-page Power BI dashboard consolidates the Sales, Customer and Product datasets into a single, filterable view of order-quantity performance by product, customer gender, price and time, giving management a centralized picture of retail demand for the reporting period.

Step 9: Output



Conclusion

This experiment successfully imported the AdventureWorks Sales, Customer and Product datasets from CSV flat files into Power BI, cleaned and typed the data in Power Query, and modeled the tables using ProductKey and CustomerKey relationships. A single-page Business Performance dashboard was built using a Total Quantity Sold KPI card together with bar, line, scatter and donut visuals, all driven by the Sum of OrderQuantity aggregation. The resulting dashboard turns three separate flat files into one connected view that supports data-driven decisions on product demand, customer segments and sales trends.

Learning Outcomes

Through this experiment, I learned:

- Data Collection and Integration — how to work with multiple related flat-file (CSV) datasets and prepare them for a single reporting model.
- Power BI Data Import — how to use Get Data → Text/CSV to bring several sources into Power BI, and how the same workflow extends to database sources such as MySQL.
- Data Cleaning and Transformation — how to use Power Query to check for missing/duplicate values, rename columns and correct data types.
- Data Modeling — how to build relationships between a transaction table and its lookup tables using ProductKey and CustomerKey, and how Power BI's automatic date hierarchy supports time-based analysis without a dedicated Calendar table.
- Aggregations in Power BI — how a single Sum of OrderQuantity aggregation can be reused across multiple visual types (card, bar, line, scatter, donut) to answer different business questions.
- Data Visualization — how to build KPI cards, bar charts, line charts, scatter charts and donut charts to represent business information visually.
- Business Analysis — how to read product-level, gender-level, time-based and price-vs-volume patterns directly from a Power BI dashboard.
- Data-Driven Decision Making — how consolidating flat files into one Power BI model turns raw transactional data into insights that support retail decision-making.